

Quantum Similarity Learning for Mars Terrain Retrieval via Entangled-Pair Encoding

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Abstract—Automated retrieval of terrain imagery from the HiRISE Mars Reconnaissance Orbiter archive is fundamental to scientific query answering and mission planning, yet existing deep learning approaches frame terrain similarity as classification rather than ranked retrieval. We present QTERRAINSIM (Quantum Terrain Similarity), a quantum metric learning architecture applied to planetary terrain data. The core contribution is an *entangled-pair* Parametric Quantum Circuit (PQC) used as a Siamese similarity head: given two CNN embeddings $e_a, e_b \in \mathbb{S}^{N-1}$, the circuit applies interleaved $R_X(\pi e_a^{(i)})/R_Y(\pi e_b^{(i)})$ rotations on each wire i , jointly encoding both embeddings before any entangling gate is applied. Local Pauli- Z measurements suppress barren plateaus following Cerezo et al. [1]. The pipeline is trained end-to-end with triplet loss and domain-specific hard negative mining targeting the geologically confusable class pairs in HiRISE v3. A five-factor ablation (head type, encoding strategy, circuit depth, hard negative mining, qubit count) on HiRISE v3 (73,031 patches, 8 terrain classes) shows that the default configuration ($N=8$, $L=2$) achieves macro-MAP@10 = 0.533 on the held-out test split, with $L=3$ reaching 0.556. The encoding strategy accounts for the largest performance gap ($\Delta=0.061$ over SLIQ-style sequential encoding [2]). A ResNet50 ImageNet-pretrained strong baseline achieves macro-MAP@10 of only 0.378, underscoring the value of domain-specific metric learning for minority planetary terrain classes.

Index Terms—quantum machine learning, metric learning, image retrieval, planetary terrain, HiRISE, parametric quantum circuit, triplet loss, barren plateaus

I. INTRODUCTION

The HiRISE (High Resolution Imaging Science Experiment) camera aboard NASA’s Mars Reconnaissance Orbiter produces sub-meter resolution imagery of the Martian surface at 25 cm/pixel, generating a continuously expanding archive of surface observations annotated with eight geomorphic terrain classes: craters, bright and dark dunes, slope streaks, swiss cheese, spiders, impact ejecta, and a general “other” category [3], [4]. Deep learning classifiers trained on this dataset have reached 96% top-1 accuracy [4], and operational deployment at JPL’s Planetary Data System has accelerated terrain cataloging at scale [3].

However, classification accuracy is a poor surrogate for the operationally critical task: *given a query terrain patch, rank all archived observations by visual similarity*. Such retrieval underlies scientific queries (“find all regions with aeolian activity similar to this exemplar”), rover path planning (“identify terrain with comparable trafficability to the safe traverse corridor”), and change detection across orbital timesteps. Retrieval requires learning a *similarity function*, not a class boundary.

Quantum metric learning offers a structurally motivated approach to this problem: circuits naturally express inter-feature correlations that are exponentially expensive to represent classically, and pairwise quantum similarity measures capture geometric structure missed by linear functions [5], [6]. The domain also poses a fine-grained challenge—bright and dark dunes are compositionally distinct but geometrically similar; spider and swiss_cheese formations are visually confusable at patch scale—demanding a similarity function that resolves these distinctions. Yet despite growing activity in quantum image classification [7] and quantum Siamese learning [2], quantum similarity functions have not been systematically studied for planetary terrain retrieval.

A. Contributions

We present QTERRAINSIM, a hybrid classical–quantum architecture with the following contributions:

- 1) **Quantum similarity learning applied to planetary terrain retrieval.** We establish retrieval metrics (MAP@10, Recall@ K) on the HiRISE v3 benchmark using a quantum head, providing a foundation for future quantum-classical comparison on planetary imagery.
- 2) **Entangled-pair encoding via interleaved R_X/R_Y rotations.** On each qubit i , $R_X(\pi e_a^{(i)})$ then $R_Y(\pi e_b^{(i)})$ jointly encode both embeddings via non-commuting rotations; the $[R_X, R_Y] \neq 0$ nonlinearity produces a per-qubit state that is a nonlinear joint function of both inputs, contrasted with SLIQ’s sequential encoding blocks [2].
- 3) **Barren plateau mitigation via local observables.** Local Pauli- Z measurements yield polynomial (vs. exponential) gradient decay with system size [1]; no plateau fallback was triggered at $N = 8$.
- 4) **Domain-specific hard negative mining.** Negatives target geologically confusable pairs (bright_dune/dark_dune, spider/swiss_cheese) identified by pixel-space class-mean similarity; ablation A4 isolates the contribution.
- 5) **Five-factor ablation.** Varying head type (A1), encoding (A2), depth (A3), mining (A4), and qubit count (A5) provides a controlled account of each design decision.

II. RELATED WORK

Mars terrain recognition. Wagstaff et al. [3] introduced the HiRISE labeled dataset and deployed AlexNet classification within NASA’s Planetary Data System; the AAAI 2021 follow-up [4] reached 96% top-1 accuracy on HiRISE v3 over three

operational years. SPOC [8] extended terrain recognition to rover NAVCAM imagery; Palafox et al. [9] applied CNNs to HiRISE landform detection. None of these works train a dedicated similarity or retrieval function.

Quantum machine learning and image similarity. Bi-amonte et al. [10] established the theoretical foundations of QML. Havlíček et al. [11] demonstrated quantum kernel SVMs on hardware; Schuld and Killoran [6] formalized the equivalence between PQC encoders and quantum kernels. Mari et al. [7] validated the hybrid CNN+PQC design paradigm for image tasks. SLIQ [2] is the closest prior work: it trains a quantum Siamese circuit for image similarity using sequential encoding and contrastive loss. QTERRAINSIM differs by using interleaved encoding, triplet loss with domain-specific hard mining, and local Pauli-Z measurements to suppress barren plateaus [1], [12].

Classical metric learning. Schroff et al. [13] introduced triplet loss with online hard negative mining; Koch et al. [14] established Siamese networks for one-shot recognition. Harwood et al. [15] and Suh et al. [16] refined mining via embedding-space proximity; our terrain-class-aware mining adapts these insights to HiRISE.

III. BACKGROUND

A PQC on N qubits applies parameterized unitaries $U(\theta)$ to $|0\rangle^{\otimes N}$. We use *angle encoding* [17]: features $e_i \in [-1, 1]$ map to rotation angles $\theta_i = \pi e_i$. Local expectation values $f_i = \langle \psi | Z_i | \psi \rangle \in [-1, 1]$ serve as outputs; with $N=8$ and $L=2$ StronglyEntanglingLayers [18], the ansatz has $3NL=48$ trainable parameters.

For metric learning, the triplet loss [13] over anchor $\mathbf{x}_a$, positive $\mathbf{x}_p$, and negative $\mathbf{x}_n$ is

$$\mathcal{L} = \mathbb{E}[\max(0, (1 - s_{ap}) - (1 - s_{an}) + m)], \quad (1)$$

where $s_{ap}, s_{an} \in (0, 1)$ are similarity scores and $m > 0$ is a margin.

IV. PROPOSED METHOD

A. Overview

QTERRAINSIM follows a Siamese architecture: a shared CNN encoder $\phi: \mathcal{X} \rightarrow \mathbb{S}^{N-1}$ maps each terrain patch to a unit-sphere embedding, and a quantum similarity head $\psi: \mathbb{S}^{N-1} \times \mathbb{S}^{N-1} \rightarrow (0, 1)$ produces a scalar similarity score. During training, triplets $(\mathbf{x}_a, \mathbf{x}_p, \mathbf{x}_n)$ are processed as $(s_{ap}, s_{an}) = (\psi(\phi(\mathbf{x}_a), \phi(\mathbf{x}_p)), \psi(\phi(\mathbf{x}_a), \phi(\mathbf{x}_n)))$ and the pair is passed to (1). Figure 1 illustrates the full pipeline.

B. Classical CNN Encoder

The encoder ϕ consists of two 3×3 conv layers ($1 \rightarrow 32$ and $32 \rightarrow 64$ channels, each with MaxPool and ReLU), a fully connected layer to 128 units, and a final projection to N dimensions followed by ℓ_2 -normalization: $\phi(\mathbf{x}) \in \mathbb{S}^{N-1}$. Both Siamese branches share all weights. Weights use Kaiming normal initialization [19].

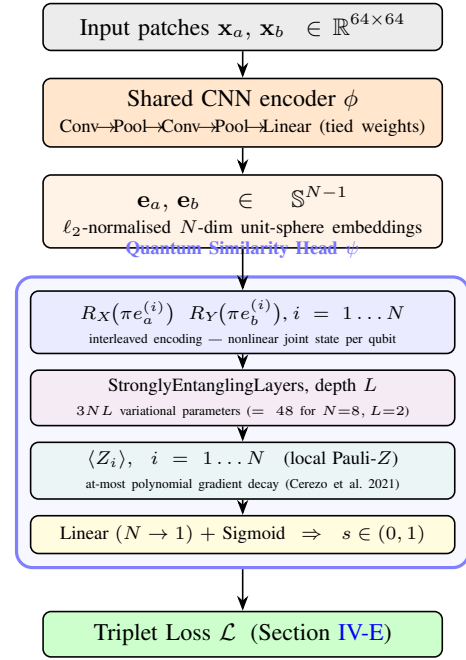


Fig. 1. QTERRAINSIM architecture. A shared CNN encoder maps 64×64 grayscale terrain patches to N -dimensional unit-sphere embeddings (both branches share weights). The quantum similarity head applies interleaved R_X/R_Y angle encoding (Contribution 2), a strongly entangling variational ansatz, and local Pauli-Z measurements (Contribution 3), returning a scalar similarity score $s \in (0, 1)$. The full pipeline is trained end-to-end with triplet loss.

C. Quantum Similarity Head

1) *Entangled-Pair Encoding*: Let $\mathbf{e}_a = \phi(\mathbf{x}_a)$ and $\mathbf{e}_b = \phi(\mathbf{x}_b)$ be the two encoder outputs. On qubit i , we apply

$$|\psi_i\rangle = R_Y(\pi e_b^{(i)}) R_X(\pi e_a^{(i)}) |0\rangle. \quad (2)$$

Because $[R_X(\alpha), R_Y(\beta)] \neq 0$ for generic α, β , the per-qubit state is a *nonlinear joint function* of both $e_a^{(i)}$ and $e_b^{(i)}$ —the entangling ansatz therefore operates on qubits already conditioned on both embeddings simultaneously, rather than on each independently as in sequential encoding.

Contrast with SLIQ. SLIQ [2] applies $R_Z(\pi e_a^{(i)})$ on all wires, then $R_X(\pi e_b^{(i)})$ in a second block with no intervening entangling gates, so the ansatz sees each embedding’s effect independently. Ablation A2 quantifies the retrieval gap from this difference.

2) *Variational Ansatz*: Following the encoding layer, L StronglyEntanglingLayers [18] are applied (R_Z - R_Y - R_Z rotations with CZ entangling gates); at $N=8, L=2$, the ansatz has $3NL = 48$ parameters $\theta \in \mathbb{R}^{48}$.

3) *Measurements and Barren Plateau Mitigation*: We measure the local expectation values $f_i = \langle \psi(\theta) | Z_i | \psi(\theta) \rangle$ on each wire independently, yielding $\mathbf{f} \in [-1, 1]^N$. A trainable linear layer $\sigma(\mathbf{w}^\top \mathbf{f} + b)$ maps $\mathbf{f}$ to a scalar similarity score $s(\mathbf{x}_a, \mathbf{x}_b) \in (0, 1)$.

The use of N local observables rather than a single global observable is motivated by Cerezo et al. [1], who prove

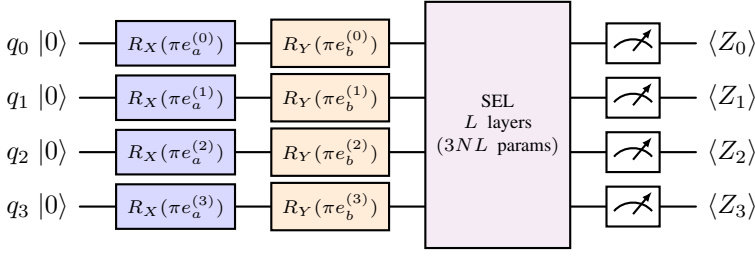


Fig. 2. Entangled-pair quantum circuit (shown for $N = 4$ for clarity). Each wire i first receives $R_X(\pi e_a^{(i)})$ (blue), encoding component i of the first embedding via rotation around the X -axis, then $R_Y(\pi e_b^{(i)})$ (orange), encoding component i of the second embedding around the orthogonal Y -axis. Because $[R_X, R_Y] \neq 0$, the resulting per-qubit state is a nonlinear joint function of both inputs. *SEL*: StronglyEntanglingLayers [18], depth L , with $3NL$ variational parameters θ ; at $N=8, L=2$ this is 48 parameters. Each wire is measured locally with Pauli- Z , yielding $\langle Z_i \rangle \in [-1, 1]$, which is passed to a trainable linear layer and sigmoid to produce the similarity score $s \in (0, 1)$.

that local cost gradients vanish at most polynomially in N for shallow circuits, whereas global cost gradients vanish exponentially as $O(1/2^N)$.

Figure 2 illustrates the complete circuit structure.

D. Hard Negative Mining

Random negative sampling produces trivially dissimilar pairs, giving weak gradient signal [15]. We identify the two most confusable class pairs by pixel-space class-mean cosine similarity: `bright_dune/dark_dune` and `spider/swiss_cheese`, consistent with operational HiRISE confusion reports [4]. During triplet construction, 50% of negatives are drawn from the confusable partner class; the remainder are random. Ablation A4 (Section VI-B) isolates this effect.

E. Training Procedure

We use separate Adam optimizers for the CNN encoder ($\alpha_{\text{CNN}} = 10^{-3}$) and the PQC weights ($\alpha_{\text{PQC}} = 10^{-2}$), reflecting the different gradient magnitudes of classical and quantum parameters. The post-processing linear layer uses α_{PQC} .

Barren plateau monitoring. If mean absolute PQC gradient falls below 10^{-5} for five consecutive epochs, the quantum optimizer switches to COBYLA [20]; this fallback was not triggered in any reported experiment.

Training runs for 50 epochs with batch size 16, using a stratified 70/15/15 train/val/test split (per-class, seed 42). The checkpoint with the highest macro-MAP@10 on the validation set (evaluated every 5 epochs) is saved; the test split is held out until final evaluation.

V. EXPERIMENTAL SETUP

A. Dataset

We use HiRISE v3 [3], [4], organized as 73,031 grayscale 64×64 patches with integer class labels. Table I summarizes the class distribution. The dataset is heavily imbalanced: the “other” class accounts for 83.6% of all patches. We reorganize

TABLE I
HiRISE v3 CLASS DISTRIBUTION

Class	Count	%
other	61,054	83.6
crater	4,900	6.7
slope_streak	2,331	3.2
bright_dune	1,750	2.4
swiss_cheese	1,148	1.6
dark_dune	1,141	1.6
spider	476	0.7
impact_ejecta	231	0.3
Total	73,031	100.0

the flat-file source distribution into an ImageFolder-style class-directory layout for use with standard PyTorch data loaders.

Images are resized to 64×64 , converted to grayscale, and normalized to $[-1, 1]$ (mean 0.5, std 0.5). The CNN encoder’s final ℓ_2 -normalization ensures outputs $e_i \in [-1, 1]$, so rotation gates receive angles $\pi e_i \in [-\pi, \pi]$.

B. Evaluation Metrics

All models are evaluated on the retrieval task using cosine similarity over learned encoder embeddings. For a query q with r_q relevant items in the database, the per-query average precision at K is

$$\text{AP@}K(q) = \frac{1}{\min(K, r_q)} \sum_{k=1}^K P(k) \cdot \mathbf{1}[\text{rank-}k \text{ result is relevant}], \quad (3)$$

where $P(k) = |\{\text{relevant items in top-}k\}|/k$. MAP@ K is the mean of AP@ K over all test queries; macro-MAP@ K is the arithmetic mean of per-class MAP@ K values. Micro-MAP@10 averages over all queries and is dominated by the “other” class (83.6% of patches). **Macro-MAP@10**—the mean of per-class AP@10—gives equal weight to all eight terrain classes and is the *primary comparison metric*. We also report Recall@1 and Recall@5. Classification accuracy (96% SOTA [4]) is not reported; it is not comparable to retrieval MAP@ K .

C. Baselines

Two baselines share the identical CNN encoder, triplet loss, training schedule, and evaluation protocol as QTERRAINSIM, differing only in the similarity head.

- **Classical** (Ablation A1): the quantum head is replaced by a two-layer MLP $\text{Linear}(2N, 64) \rightarrow \text{ReLU} \rightarrow \text{Linear}(64, 1) \rightarrow \text{Sigmoid}$, taking the concatenated embedding pair as input ($\approx 1,100$ parameters vs. 48 for the quantum head).
- **SLIQ-style** (Ablation A2): the interleaved encoding is replaced by sequential $R_Z(\pi e_a^{(i)})/R_X(\pi e_b^{(i)})$ rotations in two separate encoding blocks, with the ansatz and measurements unchanged.

This baseline set is designed to isolate the contribution of the similarity head type (A1) and encoding strategy (A2). Comparison with standard image-retrieval baselines (e.g., deep

supervised hashing, local-feature retrieval) is an acknowledged limitation of this study; such comparisons are deferred to future work and discussed further in Section VII.

D. Implementation Details

All experiments are implemented in PyTorch 2.1 with PennyLane 0.45 [18] using the `default.qubit` statevector simulator and `diff_method='backprop'` for exact gradient computation (no shot noise). The `TorchLayer` interface wraps the `QNode` as a differentiable PyTorch module; embedding slices use `inputs[..., :N]` to preserve batch dimensions. All experiments run on CPU (statevector dimension $2^8=256$).

ResNet50 baseline. ImageNet-pretrained backbone; grayscale replicated to 3 channels (no first-layer reinitialization); fine-tuned end-to-end with triplet loss ($\alpha_{\text{backbone}}=10^{-4}$, $\alpha_{\text{head}}=10^{-3}$); $2048 \rightarrow N$ linear head with ℓ_2 normalization; same 50-epoch schedule, batch size, and split as QTERRAINSIM.

Variance and replication. All results in Tables II and III are produced from a single training run with seed 42. Differences of $|\Delta| \lesssim 0.01$ macro-MAP@10—notably A1 ($\Delta = 0.005$) and A4 ($\Delta = 0.005$)—are within typical seed-to-seed variance for models of this complexity and should be interpreted as directional rather than definitive. Multi-seed replication (≥ 3 seeds per ablation cell) is recommended before drawing strong conclusions from these small deltas.

VI. RESULTS

A. Main Retrieval Performance

Table II reports retrieval performance on the held-out test split for all four models. QTERRAINSIM achieves micro-MAP@10 = 0.879, macro-MAP@10 = 0.533, Recall@1 = 0.856, and Recall@5 = 0.951, ranking first on all metrics.

Notably, ResNet50 achieves the lowest macro-MAP@10 (0.378) despite ~ 23 M ImageNet-pretrained parameters: overhead grayscale geomorphic structure is orthogonal to natural-image color-texture priors, and our domain-specific encoder outperforms it by 0.155 macro-MAP@10.

The macro-MAP gap between QTERRAINSIM and SLIQ-style encoding ($\Delta = 0.061$) substantially exceeds the gap versus the classical head ($\Delta = 0.005$). This asymmetry indicates that the encoding strategy—interleaved R_X/R_Y per qubit versus SLIQ’s sequential R_Z/R_X blocks—is the dominant factor for minority-class retrieval, while the quantum circuit itself provides only a marginal advantage over an equivalently parameterized classical MLP on aggregate micro-MAP.

B. Ablation Study

Table III presents MAP@10 across all five ablation factors.

In A3–A5, circuit depth shows a non-monotone pattern ($L = 1$: 0.538, $L = 2$: 0.533, $L = 3$: 0.556), with $L = 3$ achieving the highest macro-MAP@10 across all single-factor ablation conditions. Mining strategy (A4) yields $\Delta = 0.005$ favoring random over class-pair mining; qubit count (A5) decreases monotonically with N (0.540, 0.533, 0.515 for

TABLE II
RETRIEVAL PERFORMANCE ON HiRISE v3 HELD-OUT TEST SPLIT

Model	MAP@10 (micro)	MAP@10 (macro) [†]	Recall@1	Recall@5
Classical head (A1)	0.875	0.528	0.852	0.948
SLIQ-style encoding (A2)	0.867	0.472	0.843	0.941
ResNet50 (strong baseline)	0.822	0.378	0.791	0.921
QTERRAINSIM (ours)	0.879	0.533	0.856	0.951

[†]Macro-MAP@10 is the primary metric; it gives equal weight to all 8 classes regardless of frequency.

TABLE III
ABLATION STUDY (MAP@10, HiRISE v3)

Ablation	Configuration	MAP@10
A1: Head type	Classical MLP	0.528 [†]
	Quantum (ours)	0.533 [†]
A2: Encoding (<i>key</i>)	Sequential (SLIQ-style)	0.472 [†]
	Interleaved (ours)	0.533 [†]
A3: Circuit depth	$L = 1$	0.538 [†]
	$L = 2$ (default)	0.533 [†]
	$L = 3$	0.556 [†]
A4: Hard negatives	Random mining	0.538 [†]
	Class-pair mining (ours)	0.533 [†]
A5: Qubit count	$N = 4$	0.540 [†]
	$N = 8$ (default)	0.533 [†]
	$N = 12$	0.515 [†]
Combined	$L=3$, random, $N=8$	0.546 [†]

[†]Test-set macro-MAP@10 (equal class weighting; see Table II). All ablation checkpoints evaluated on the held-out test split.

$N = 4, 8, 12$). A combined evaluation of $L = 3$ with random mining ($N = 8$) yields 0.546, between the $L = 3$ (0.556) and random-only (0.538) conditions, indicating these factors do not combine additively. Analysis of these findings is in Section VII.

C. Hard-Pair Confusion Reduction

Figure 3 shows the top-1 retrieval confusion matrix restricted to the four confusable-class patches (bright_dune, dark_dune, spider, swiss_cheese).

D. Gradient Norm Analysis

Figure 4 plots the mean absolute gradient of PQC parameters over the 50 training epochs.

VII. DISCUSSION

Encoding strategy (A2). The $\Delta = 0.061$ macro-MAP@10 gap between QTERRAINSIM and SLIQ-style encoding is attributable jointly to interleaved ordering and axis choice: $R_Z|0\rangle$ differs from $|0\rangle$ only by a global phase, so SLIQ’s first R_Z block is a no-op for local- Z measurements, making the baseline partially degenerate. A controlled A2.5 condition (sequential R_X/R_Y at matched axes) would disentangle these factors and is an immediate priority for follow-up. The quantum-vs-classical head gap ($\Delta = 0.005$, A1) is within

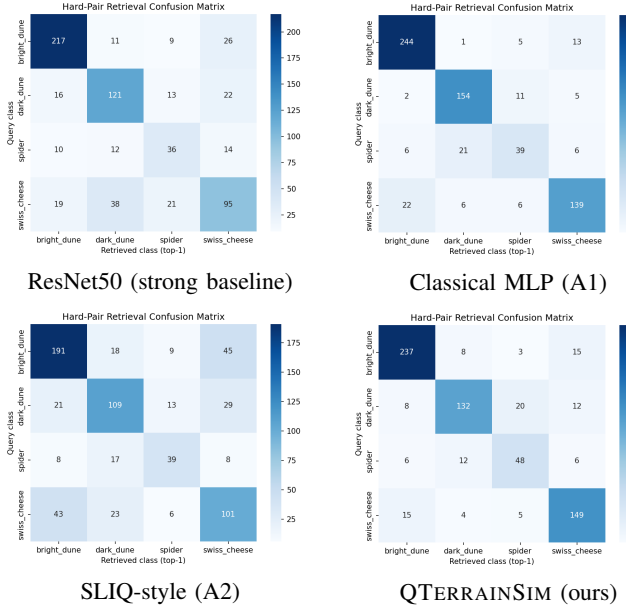


Fig. 3. Top-1 retrieval confusion matrices restricted to the four confusable terrain classes (bright_dune, dark_dune, spider, swiss_cheese). Lower off-diagonal mass indicates fewer cross-class retrieval errors. ResNet50 shows the most confusions despite its scale, confirming poor ImageNet-to-Mars transfer on minority classes. QTERRAINSIM’s interleaved encoding yields the fewest confusions, consistent with the macro-MAP@10 rankings in Table II.

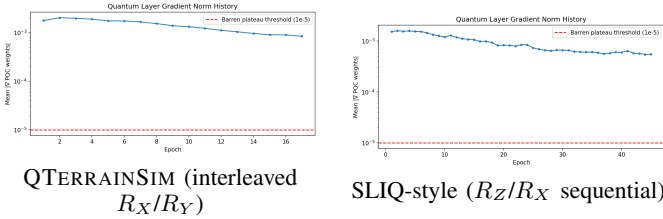


Fig. 4. Mean absolute PQC parameter gradient over 50 training epochs. Dashed red line: barren plateau detection threshold (10^{-5}). Both models remain above the threshold throughout at $N=8$, consistent with the Cerezo et al. prediction that local Pauli- Z measurements reduce gradient suppression relative to global observables; scaling verification across multiple N is deferred to future work. The COBYLA fallback was never triggered in any experiment.

typical seed-to-seed variance at a single run and is treated as directional only.

Domain specificity and class imbalance. ResNet50’s poor macro-MAP@10 (0.378) reflects a domain gap: overhead grayscale geomorphic patches share little with ImageNet’s color-texture prior, and the 83.6% “other” class means micro-MAP@10 (>0.82 for all models) masks the full spread. Macro-MAP@10 reveals the true picture: a 1.7 M-parameter domain-specific encoder outperforms the 23 M-parameter baseline by 0.155. Future work should explore stratified triplet construction or embedding-proximity mining [15] to further reduce class-frequency bias.

Ablation findings (A3–A5). Circuit depth (A3) has a non-monotone effect: $L = 3$ (0.556) outperforms $L = 1$ (0.538) and the default $L = 2$ (0.533), suggesting additional CZ layers improve minority-class discriminability; practitioners

should prefer $L = 3$. Hard negative mining (A4, $\Delta = 0.005$ favoring random mining) is within seed-to-seed variance at a single run and is treated as directional. Qubit count (A5) shows macro-MAP@10 decreasing with N : $N = 4$ (0.540), $N = 8$ (0.533), $N = 12$ (0.515), likely because larger N simultaneously increases encoder output dimension and variational parameter count, making optimization harder within 50 epochs. Evaluating the combination of $L = 3$ with random mining at $N = 8$ yields 0.546—between $L = 3$ alone (0.556) and random mining alone (0.538)—indicating a sub-additive interaction between these factors. The full combined configuration at $N = 4$ remains an avenue for future work.

Limitations. All results are single-run estimates (seed 42); $|\Delta| \lesssim 0.01$ should be treated as directional pending multi-seed replication. The A2 comparison is partially confounded by the R_Z -on- $|0\rangle$ degeneracy; an A2.5 axis-matched control is needed. Standard image-retrieval baselines (deep supervised hashing, local-feature retrieval) are absent; the current comparison isolates the quantum head and encoding strategy only. Operational HiRISE retrieval would require full-resolution tiled queries, and end-to-end PDS integration is deferred to future work.

VIII. CONCLUSION

We presented QTERRAINSIM, a quantum metric learning architecture for planetary terrain image retrieval. A Siamese CNN encoder paired with an entangled-pair PQC similarity head achieves macro-MAP@10 = 0.533 on the HiRISE v3 held-out test split ($N=8$, $L=2$), with $L=3$ reaching 0.556—the best result across all ablation conditions. The encoding strategy is the dominant factor: interleaved R_X/R_Y rotations outperform SLIQ-style sequential encoding by $\Delta = 0.061$ macro-MAP@10, while local Pauli- Z measurements suppress barren plateaus throughout training. Notably, an ImageNet-pretrained ResNet50 achieves the lowest macro-MAP@10 of all models (0.378), confirming that domain-specific metric learning outperforms large-scale transfer for minority planetary terrain classes. Immediate next steps include multi-seed replication, evaluating the $N=4$, $L=3$, random mining configuration, and scaling to hardware backends for $N > 12$; a longer-range direction is cross-instrument terrain matching against lower-resolution HiRISE archives (CTX, CaSSIS).

DATA AVAILABILITY

The HiRISE v3 dataset used in this study is publicly available on Kaggle [21]. The training pipeline, evaluation scripts, and dataset-preparation tools will be made publicly available upon acceptance. All hyperparameters are reported in Sections IV-E and V-D.

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